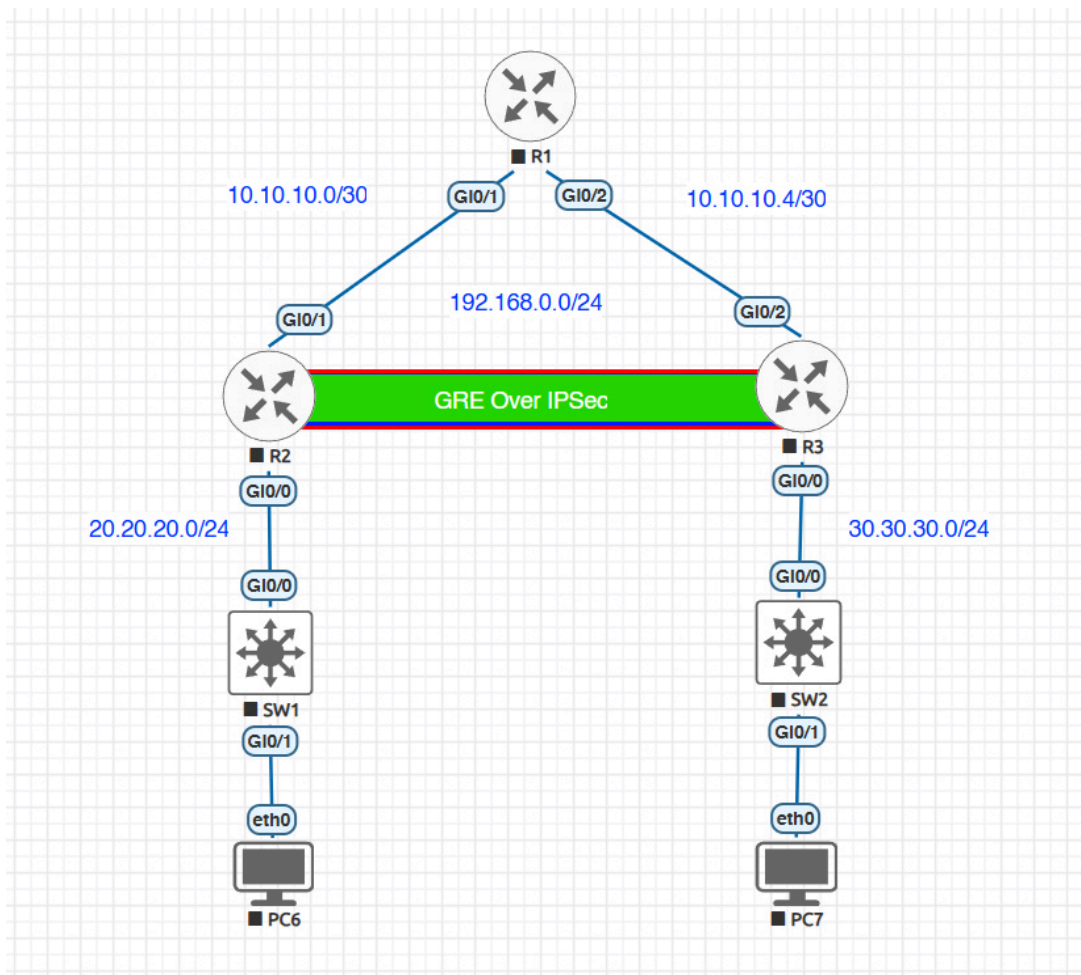


Project Plan: GRE over IPsec Implementation (CCNP Enterprise Lab)

Objective: To design and implement a secure, scalable site-to-site VPN solution utilizing GRE over IPsec with multi-protocol routing (OSPF and EIGRP) and NAT optimization.



I. Executive Summary

This document outlines the professional deployment strategy for connecting branch offices via a virtual tunnel. The design separates the "Underlay" (ISP/Public routing) from the "Overlay" (Private/Tunnel routing) to ensure maximum stability and security. Key features include dynamic routing over the tunnel and fragmentation prevention via MTU/MSS tuning.

II. Deployment Phase 1: The Underlay (Physical Layer)

The goal is to establish basic IP reachability between the public-facing interfaces of the edge routers.

- **Interface Configuration:** Assign static Public IPs to the WAN interfaces and Private IPs to the LAN interfaces.
- **WAN Routing (OSPF):** Implement OSPF to simulate the ISP backbone or public internet.
 - *Constraint:* Advertise only WAN/Public subnets. **DO NOT advertise LAN networks into the public OSPF process.**
- **Verification:** Execute ping [Remote_Public_IP]. Reachability must be 100% before proceeding.

III. Deployment Phase 2: The Overlay (GRE Tunneling)

Establishing the logical "Point-to-Point" link between sites.

```
interface Tunnel0
ip address [Tunnel_Private_IP] [Mask]
tunnel source [Local_Public_IP]
tunnel destination [Remote_Public_IP]
```

- **LAN Routing (EIGRP):** Configure EIGRP to handle internal traffic.
 - Advertise Local LAN subnets.
 - Advertise the Tunnel IP subnet.
 - Ensure EIGRP is active on the Tunnel0 interface.
- **Verification:** Check for EIGRP neighbor up status on the tunnel and verify the remote LAN appears in the routing table (show ip route eigrp).

IV. Deployment Phase 3: The Security Layer (IPsec)

Securing the GRE "straw" with a cryptographic shield.

Component	Description	Command Context
IKE Phase 1	Negotiates management session (ISAKMP)	crypto isakmp policy
IKE Phase 2	Negotiates data encryption parameters	crypto ipsec transform-set
Protection	Binds IPsec to the Tunnel interface	tunnel protection ipsec profile

Verification: show crypto isakmp sa should display **QM_IDLE**.

V. Deployment Phase 4: NAT & Path Optimization

Ensuring LAN users can reach the internet while maintaining VPN integrity.

- **NAT Exemption (Policy NAT):** Create an Access Control List that prevents the router from NAT-ing traffic destined for the remote branch.

```
ip access-list extended NAT_ACL
deny ip [Local_LAN] [Remote_LAN]
permit ip [Local_LAN] any
```

- **Application:** ip nat inside source list NAT_ACL interface [WAN_INT] overload

VI. Deployment Phase 5: MTU and MSS Tuning

To prevent packet drops due to encapsulation overhead (GRE + IPsec), the following adjustments are mandatory on the **Tunnel0** interface:

Command	Value	Purpose
ip mtu	1400	Adjusts the maximum transmission unit for the tunnel.
ip tcp adjust-mss	1360	Ensures TCP headers fit within the MTU without fragmentation.

VII. Troubleshooting & Maintenance Commands

- **show ip eigrp neighbors**: Confirm internal routing stability.
- **show crypto ipsec sa**: Verify packet encryption/decryption counters.
- **show ip route**: Confirm administrative distance (Static vs OSPF vs EIGRP).

Running Config

R1

```
hostname R1
!
interface GigabitEthernet0/1
 ip address 10.10.10.1 255.255.255.252
 duplex full
 speed 1000
 media-type rj45
!
interface GigabitEthernet0/2
 ip address 10.10.10.5 255.255.255.252
 duplex full
 speed 1000
 media-type rj45
!
router ospf 1
 network 10.10.10.0 0.0.0.7 area 0
!
```

R2

```
hostname R2
!
crypto isakmp policy 10
 encr aes
 hash md5
 authentication pre-share
 group 2
crypto isakmp key cisco address 0.0.0.0
!
crypto ipsec transform-set SET1 esp-aes esp-sha-hmac
 mode tunnel
!
crypto ipsec profile GREIPSEC
 set transform-set SET1
!
interface Tunnel0
 ip address 192.168.0.1 255.255.255.0
 tunnel source GigabitEthernet0/1
 tunnel destination 10.10.10.6
 tunnel protection ipsec profile GREIPSEC
!
interface GigabitEthernet0/0
 ip address 20.20.20.1 255.255.255.0
 ip nat inside
 ip virtual-reassembly in
 duplex full
```

```
speed 1000
media-type rj45
!
interface GigabitEthernet0/1
ip address 10.10.10.2 255.255.255.252
ip nat outside
ip virtual-reassembly in
duplex full
speed 1000
media-type rj45
!
router eigrp 100
network 20.0.0.0
network 192.168.0.0
passive-interface GigabitEthernet0/0
!
router ospf 1
network 10.10.10.0 0.0.0.3 area 0
!
ip nat inside source list NAT_ACL interface GigabitEthernet0/1 overload
!
ip access-list extended NAT_ACL
deny ip 20.20.20.0 0.0.0.255 30.30.30.0 0.0.0.255
permit ip 20.20.20.0 0.0.0.255 any
!
```

R3

```
!
version 15.6
service timestamps debug datetime msec
hostname R3
!
crypto isakmp policy 10
encr aes
hash md5
authentication pre-share
group 2
crypto isakmp key cisco address 0.0.0.0
!
crypto ipsec transform-set SET1 esp-aes esp-sha-hmac
mode tunnel
!
crypto ipsec profile GREIPSEC
set transform-set SET1
!
interface Tunnel0
ip address 192.168.0.2 255.255.255.0
tunnel source GigabitEthernet0/2
```

```
tunnel destination 10.10.10.2
tunnel protection ipsec profile GREIPSEC
!
interface GigabitEthernet0/0
ip address 30.30.30.1 255.255.255.0
ip nat inside
ip virtual-reassembly in
duplex full
speed 1000
media-type rj45
!
interface GigabitEthernet0/2
ip address 10.10.10.6 255.255.255.252
ip nat outside
ip virtual-reassembly in
duplex full
speed 1000
media-type rj45
!
router eigrp 100
network 30.0.0.0
network 192.168.0.0
passive-interface GigabitEthernet0/0
!
router ospf 1
network 10.10.10.4 0.0.0.3 area 0
!
ip nat inside source list NAT_ACL interface GigabitEthernet0/2 overload
!
ip access-list extended NAT_ACL
deny ip 30.30.30.0 0.0.0.255 20.20.20.0 0.0.0.255
permit ip 30.30.30.0 0.0.0.255 any
!
```